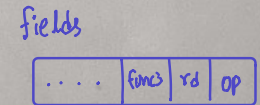
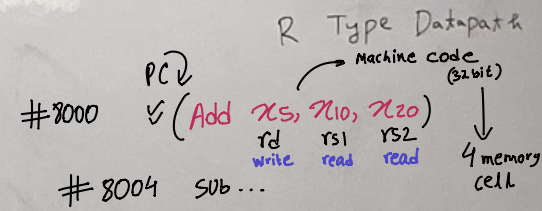
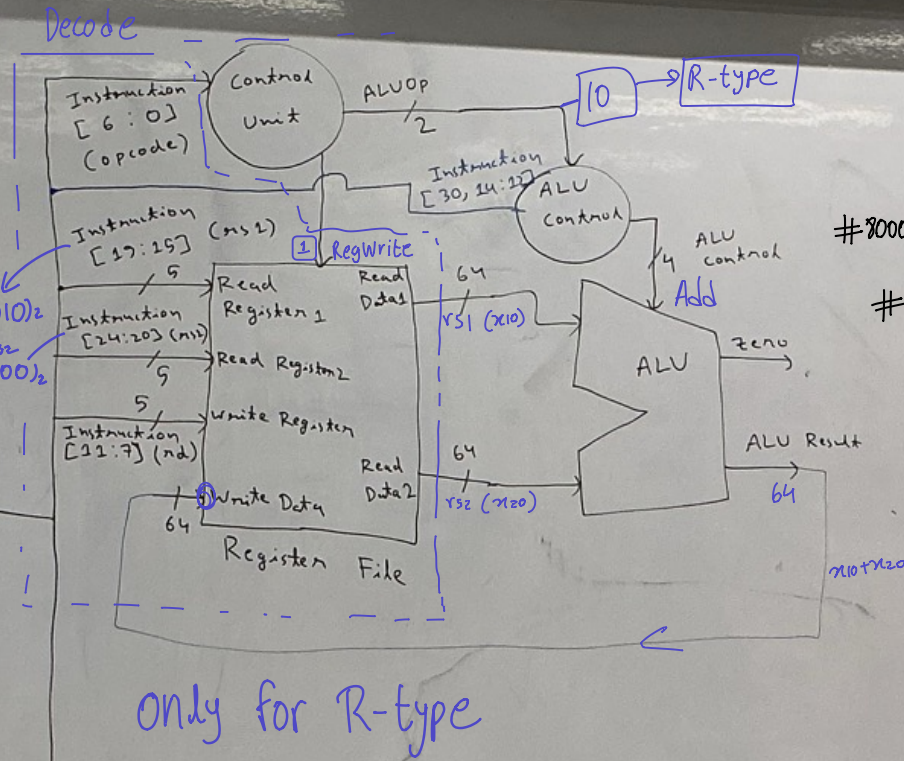
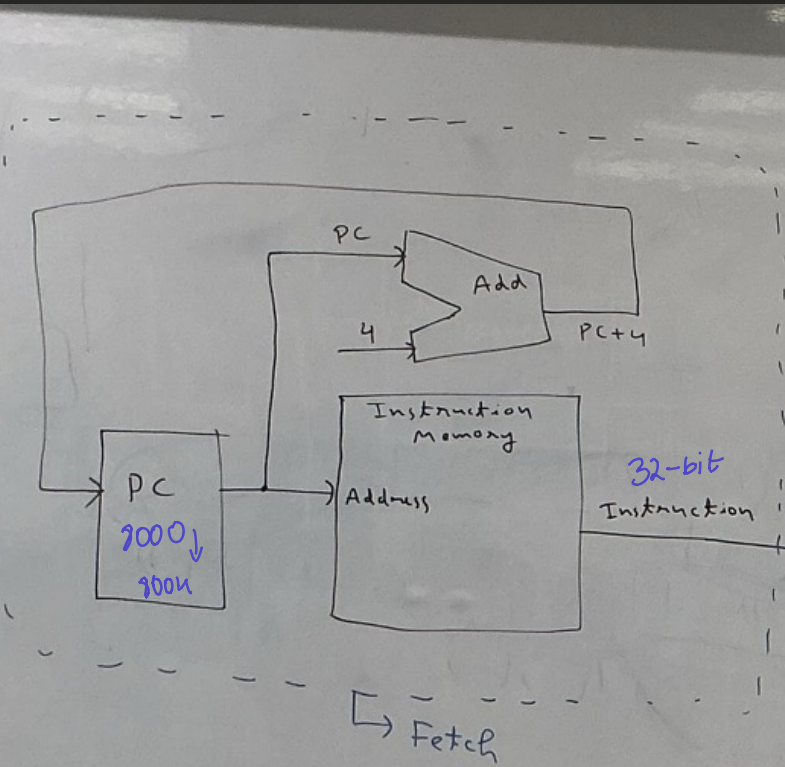


Chapter 4 10th August (look at slides too)



Only draw what is needed for the specific type of instruction

The diagram illustrates the internal structure of a MIPS processor, divided into two main stages: **Fetch** and **Decode**.

Fetch Stage:

- The **PC** (Program Counter) is initially 8004. It is incremented by 4 (PC+4) to determine the next instruction address.
- The **Instruction Memory** is accessed using the PC value to retrieve the **Instruction** (32-bit).

Decode Stage:

- The instruction is split into fields: **OpCode** (6 bits), **Register 1** (5 bits), **Register 2** (5 bits), **Write Register** (5 bits), and **Immediate** (16 bits).
- The **Control Unit** interprets the **OpCode** to determine the **ALUOP** (ALU operation) and **RegWrite** (whether to write to registers).
- The **ALU** (Arithmetic Logic Unit) performs the operation on the values from **Register 1** and **Register 2** (or the **Immediate** value, depending on the operation). It also generates a **Zero** flag.
- The **Memory Address** is calculated by adding the **Register 1** value and the **Immediate** value (offset).
- The **Data Memory** is accessed using the **Memory Address** to **Read Data** (for load instructions) or **Write Data** (for store instructions).

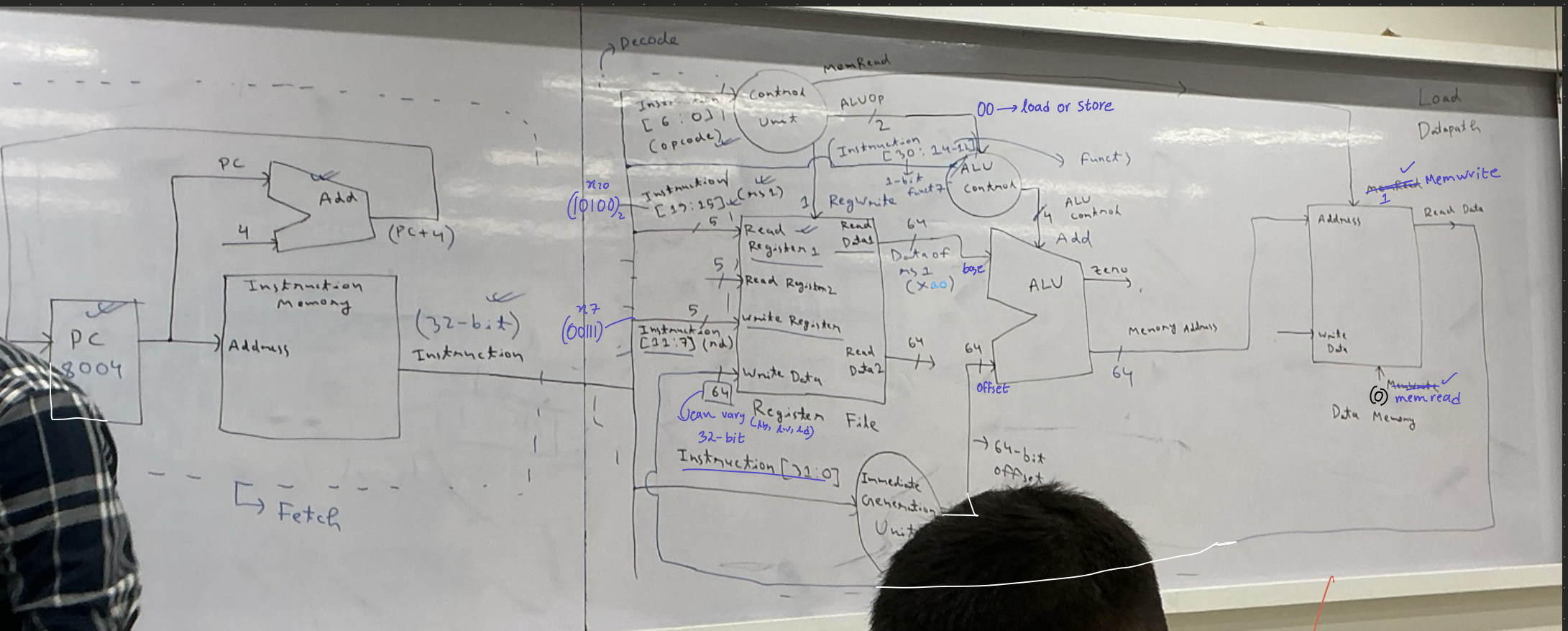
Annotations and Examples:

- Example instruction: $00100101010101010101010101010101_2$ (hex 0x10101010).
- OpCode: 00 (load or store).
- Register 1: 01010 (R2).
- Register 2: 01010 (R2).
- Write Register: 01010 (R2).
- Immediate: 0101010101010101 (0x10101010).

Memory address = $\frac{\text{base}}{64 \text{ bit}} + \frac{\text{offset}}{12 \text{ bit}}$

ALU

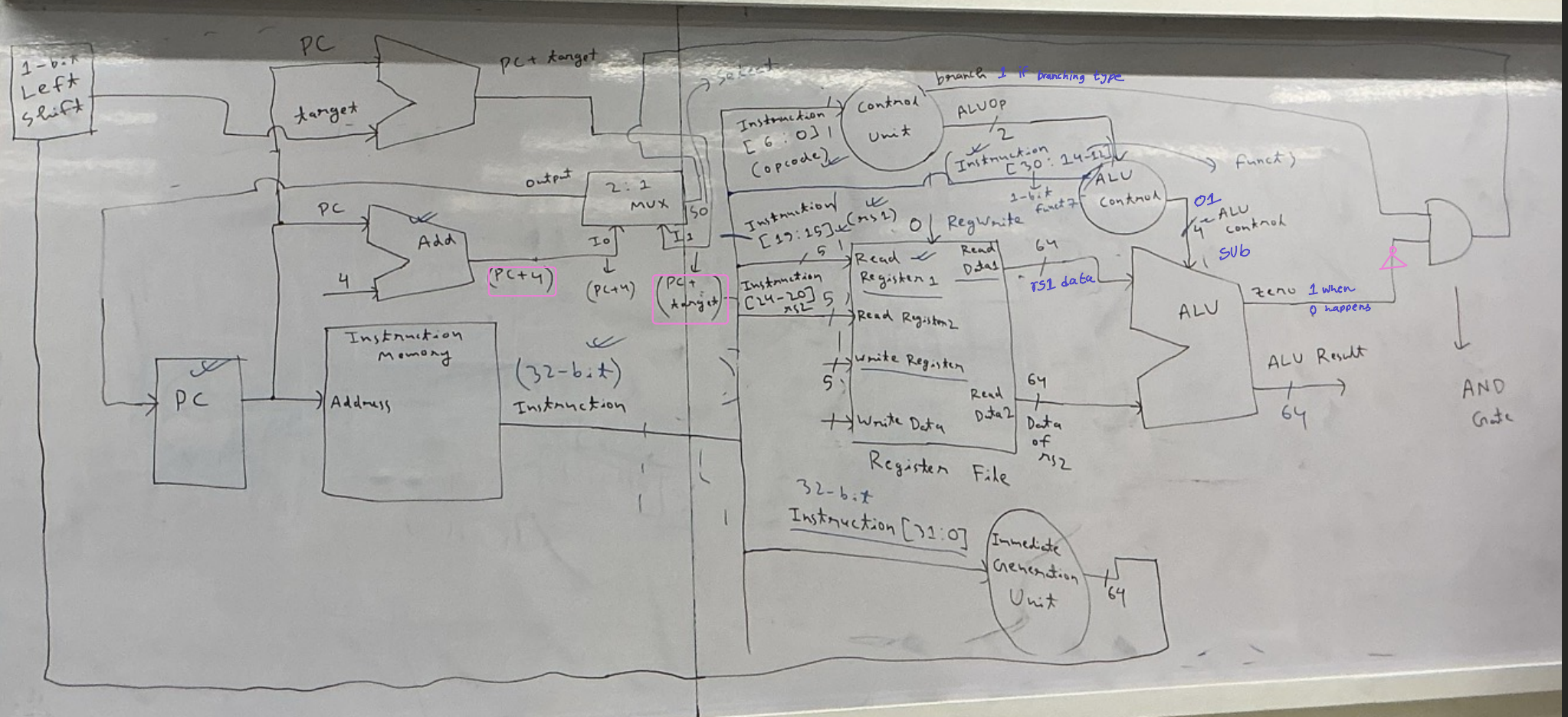
Store Datapath



sd r7, 24(r20)

Control unit line drawn only
when significant output
(1)

Branch Datapath (specifically beq)



#8000 beq r7, r20, Exit
 #8004 sub
 #8008 Addi
 Exit:
 #80012

PC+4
 PC+target
 target

Jumping offset: 6 (actual 12)

01100 → 12

00110 → 6

left shift 1ce
 01100

For bne,
 Add a not gate

Universal gates (NAND, NOR)
 MUX changes can come
 (4:1 MUX)

BGE

